Status	Finished
Started	Wednesday, 21 October 2026, 5:05 PM
Completed	Wednesday, 21 October 2026, 5:11 PM
Duration	6 mins 29 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Linear search ✓
- b. Bubble search
- c. Quick search
- d. Merge search

Question **2**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Sorting elements in a list
- b. Deleting elements from a list
- c. Determining whether an element is present in a list ✓
- d. Inserting elements into a list

Question **3**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search backtracks to find duplicate elements
- b. The search continues until the end of the list
- c. The search stops immediately ✓
- d. The search starts over from the beginning

Question **4**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the right sublist
b. The search continues in the left sublist
c. The search ends successfully ✓
d. The list is sorted

Question **5**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the left sublist ✓
b. In the entire list
c. At the beginning of the list
d. In the right sublist

Question **6**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n
b. n/2
c. n-1
d. n ✓

Question **7**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(log n)
b. O(n log n)
c. O(1)
d. O👎 ✓

Question **8**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search does not require dividing the list
- b. Binary search has a lower time complexity on large, sorted lists ✓
- c. Binary search can find multiple instances of the target element
- d. Binary search works on unsorted lists

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O👎
- b. O(1)
- c. O(n log n)
- d. O(log n) ✓

Question **10**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is the last element in the array
- b. When the item is not the array at all
- c. Item is the last element in the array or item is not there at all
- d. When the item is somewhere in the middle of the array ✓

Question **11**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(log n)
- b. O(1) ✓
- c. O(n log n)
- d. O👎

Question **12**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the list is sorted
- b. When the list contains duplicate elements
- c. When the search algorithm finishes
- d. When the element is found in the list ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By comparing each element sequentially
- b. By using a hash function
- c. By starting from the first element
- d. By dividing the list length by two ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Checking each element sequentially ✓
- b. Dividing the list in half repeatedly
- c. Skipping every second element
- d. Sorting the list before searching

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list contains duplicate elements
- b. When the list is sorted ✓
- c. When the list is very small
- d. When the list is unsorted